

EDITORIAL & MATHEMATICAL REVIEW

Level 7, Section 7.1 - Numbers, sets, arithmetic, real numbers and countability

Material: level7.1.pdf and supplied LyX source

Review scope: Mathematical correctness, logical completeness, clarity and copy-editing

Assumed background: Levels 1-6 as described: A-level Mathematics and Further Mathematics, with the necessary real analysis and linear algebra

Excluded: LaTeX/editor artifacts, hyperlink boxes and purely renderer-generated spacing

Review date: 30 July 2026

Overall assessment

The notes have a clear narrative route and many useful motivating examples. Most of the recommended changes are local. The main revision priority is a compact group of definitions, theorem statements and proof steps where the present wording changes the mathematics. Correcting those points will also make the surrounding informal explanations more dependable for a reader learning the material from first principles.

How the labels are used. "Correction" changes a mathematical statement, definition or deduction; "Clarification" makes a valid idea precise enough to use; "Copy edit" addresses wording or typography; and "Fact check" marks a time-sensitive external claim.

Priority correction register

- Correct the truth table on p. 3: when A is true and B is false, $A \wedge B$ is false.
- Repair the foundational definitions on pp. 5-8: function graphs use ordered pairs, the image and pre-image variables must be consistent, the ordered-pair definition is circular, the indicator-function definition changes variables, and four entries in the relations table are reversed.
- Recast the Peano and induction material on pp. 8-10. The successor axioms need the non-successor axiom, the treatment of 0 must match the chosen convention for $\mathbb{N}$, strong induction needs a base case, and WOP does imply strong induction on $\mathbb{N}$.
- Change the definition of composite number on p. 12 so that 1 is neither prime nor composite.
- Use congruence notation consistently on pp. 15-18. 'b is the remainder when a is divided by n' is not equivalent to $a \equiv b \pmod{n}$, and the addition and multiplication rules need either another reduction or congruence signs.
- Add 'distinct' to the primes p and q in the formula $\phi(pq) = (p-1)(q-1)$ on p. 17.
- Correct the ordered-field and completeness axioms on p. 20. A strict order is asymmetric, and a supremum need not belong to the set.

- Replace the square-root-of-2 argument on p. 22: the second inequality has the wrong direction, and the set being used switches silently from rational to real elements.
- Rewrite the convergence definition and its negation on pp. 23-24 so that the proposed limit is fixed before ϵ is chosen and all quantifiers are explicit. Also replace the reciprocal definition of divergence to $+\infty$.
- Complete the Liouville-number proof on p. 28 by comparing the lower bound $10^{-(d \cdot n!)} with the Lipschitz upper bound $2K \cdot 10^{-(n+1)!}$. Non-zero multiples of a shrinking quantity can certainly converge to zero.$
- Repair the cardinality arguments on pp. 30-32: handle decimal/binary non-uniqueness, replace the incorrect tangent formula, and reverse the stated injection test for proving uncountability.
- Replace the decimal-interleaving and Cantor-Schroeder-Bernstein passages on pp. 31-33 with versions that handle exceptional expansions and partial inverses explicitly.

Page-by-page review

Page numbers below are the PDF page numbers; they coincide with the printed footer numbers in this file. Quoted cues are deliberately short so that each point can be found quickly in either the PDF or the source.

Page 1

CORRECTION Proof by contradiction. Cue: "*proofs by contradiction ... are a sequence of false statements*".

A proof by contradiction consists of valid deductions made under a temporary assumption. The contradiction shows that the assumption is false; the intermediate statements are not simply a sequence of false statements. **Suggested revision:** Use: "In a proof by contradiction we assume the negation of the desired conclusion, make valid deductions, and derive an impossibility. Therefore the temporary assumption was false."

CORRECTION Definition of the rationals. Cue: "*an element of $\mathbb{N}$ divided by an element of $\mathbb{Z}$* ". With $\mathbb{N} = \{1, 2, 3, \dots\}$, this description omits 0 and obscures the non-zero-denominator condition. **Suggested revision:** Use $\mathbb{Q} = \{a/b : a, b \in \mathbb{Z} \text{ and } b \neq 0\}$.

Page 2

CORRECTION Algebra in the parity proof. Cue: " $(2k+1)^2 = 2(2k^2+2)+1$ ". The middle term $4k$ has been lost. **Suggested revision:** Replace it with $(2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k)+1$.

CLARIFICATION Domain of the divisibility statement. Cue: "*If 2 divides n^2 then n is even*". Divisibility and parity require n to be an integer. **Suggested revision:** State: "For $n \in \mathbb{Z}$, if $2 \mid n^2$, then $2 \mid n$."

CLARIFICATION Quantifier syntax. Cue: " $\forall x \in \mathbb{R} (\exists y \in \mathbb{R} \text{ -- s.t. } y < x)$ ". The phrase 'such that' should not sit between a quantifier and its predicate, and the punctuation makes the scope harder to see. **Suggested revision:** Write $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $y < x$, and similarly for the reversed quantifiers.

Page 3

CORRECTION Diagnosis of the false proof. Cue: "What I have really proven is that the smallest real number does not exist". The displayed argument only treats the case $r < 1$; it has not ruled out $r = 1$ or $r > 1$ and therefore has not established non-existence. **Suggested revision:** A direct diagnosis is: the opening sentence assumes an object whose existence has not been proved. To prove that no least positive real exists, if $r > 0$ then $r/2$ is positive and smaller than r .

CORRECTION Truth-table entry. Cue: "row $A=True, B=False$ has $A \wedge B=True$ ". A conjunction is true only when both component statements are true. **Suggested revision:** Change that entry to False. The corrected row is: True, False, True, False, False, False, True, False.

CLARIFICATION Extensional equality of sets. Cue: "sets ... are equal if they have the same elements". The foundational principle is a biconditional, not only a sufficient condition. **Suggested revision:** Use: "Two sets are equal if and only if they have exactly the same elements."

Page 4

CLARIFICATION Prose version of De Morgan's laws. Cue: "(Not in B or C) is the same as (Not in B and Not in C)". In ordinary English, 'not in B or C' can mean $\neg B \vee \neg C$, whereas the formula requires $\neg(B \cup C)$, meaning neither in B nor in C. **Suggested revision:** Say explicitly: "not in B $\cup$ C" means "in neither B nor C", while "not in B $\cap$ C" means "not in at least one of B and C".

CLARIFICATION Finite and infinite indexed families. Cue: " $A_1, \dots, A_n$... with n possibly infinite". The symbol n denotes a natural-number endpoint, so this notation describes a finite family. **Suggested revision:** Use $A_1, \dots, A_n$ for finite families and $(A_i)_{i \in I}$, with $\bigcap_{i \in I} A_i$ and $\bigcup_{i \in I} A_i$, for an arbitrary family.

CORRECTION Definition of finite set. Cue: "A set is defined to be finite if its size is a natural number". This is circular unless 'size' has already been defined, and it excludes the empty set under the notes' convention $N = \{1, 2, \dots\}$. **Suggested revision:** Define A to be finite if A is bijective with $\{1, \dots, n\}$ for some $n \in \mathbb{N}$, or $A = \emptyset$ (equivalently allow $n=0$).

CLARIFICATION Russell's paradox and set formation. Cue: "we should only construct sets from finite sets or known sets". The intended restriction is the separation principle - forming subsets of an already available set - rather than a restriction to finite starting sets. **Suggested revision:** Say: "Unrestricted comprehension is not allowed. Given an existing set A , we may form $\{x \in A : P(x)\}$; the unrestricted collection $\{x : P(x)\}$ need not be a set."

Page 5

CORRECTION Graph of a function. Cue: "ordered pairs of the form $\{a, f(a)\}$ ". A graph contains ordered pairs, not unordered two-element sets. **Suggested revision:** Replace $\{a, f(a)\}$ with $(a, f(a))$, so the graph is $\{(a, f(a)) : a \in A\}$.

CORRECTION Image and pre-image variables. Cue: "the image of an element a is $M \dots f(x)=B$ ". M and B do not match the previously introduced a, b and f . **Suggested revision:** Use: "The image of $a \in A$ is $f(a)$. The pre-image of $b \in B$ is $f^{-1}(\{b\}) = \{x \in A : f(x) = b\}$."

CLARIFICATION Pre-image as a set. Cue: "the pre-image of 4 is 2 and -2". A pre-image of a point is a set, even when it has one or two members. **Suggested revision:** Write $f^{-1}(\{4\}) = \{-2, 2\}$.

CORRECTION Ordered-pair definition. Cue: "An ordered pair is a set (a, b) , or $\{(a, b), a\}$ ". This is circular because it uses (a, b) in its own set-theoretic definition. **Suggested revision:** If a construction is wanted, use Kuratowski's definition $(a, b) := \{\{a\}, \{a, b\}\}$; otherwise simply treat ordered pairs as already known.

Page 6

CLARIFICATION Finite sets of equal size. Cue: "if the sets are finite then any function from A to B is injective if and only if it is surjective". This equivalence requires $|A| = |B|$. For finite sets of different sizes it is false.

Suggested revision: Insert: "If A and B are finite and $|A| = |B|$, then $f: A \rightarrow B$ is injective if and only if it is surjective."

CLARIFICATION Domain and codomain. Cue: "specify the domain and range". Surjectivity is defined relative to a codomain; 'range' is often used for the actual image, which would make every function surjective onto its range. **Suggested revision:** Use "domain and codomain" here, reserving "image/range" for $f(A)$.

CORRECTION Indicator-function definition. Cue: "For a set X and a subset $Y \dots$ denoted $i_A \dots 1$ if A is in Y ". The notation changes from Y to A and uses a set where the input variable should appear. **Suggested revision:** Use $i_A: X \rightarrow \{0, 1\}$, defined by $i_A(x) = 1$ if $x \in A$ and $i_A(x) = 0$ otherwise.

Page 7

CLARIFICATION Left inverse of an injection. Cue: "if f is injective then there is a left inverse". For $f: A \rightarrow B$, extending the inverse from $f(A)$ to all of B requires an element of A . It fails when $A = \emptyset$ and $B \neq \emptyset$. **Suggested revision:** Add "provided A is non-empty" (or deal separately with the empty case) and send every $b \notin f(A)$ to one fixed $a_0 \in A$.

CLARIFICATION Right inverse of a surjection. Cue: "for every element ... pick any element ... that maps to it". For arbitrary sets, choosing one point in every fibre is an instance of the axiom of choice. It is harmless for finite sets but should be acknowledged in a first-principles account. **Suggested revision:** Either restrict the assertion to finite sets, or add "assuming the axiom of choice".

CORRECTION Right-inverse explanation. Cue: "everything in the domain of B gets hit ... and thus f gets hit". The domains and images are interchanged in this sentence. **Suggested revision:** Use: if $f \circ g = \text{id}_B$, then for every $b \in B$ we have $b = f(g(b))$, so b lies in $f(A)$; hence f is surjective.

Page 8

CORRECTION Relations table - example 3. Cue: " aRb if $a \neq b$ ". This relation is symmetric and not reflexive. The table records the reverse answers for those two properties. **Suggested revision:** Example 3 should read: symmetric Yes; reflexive No; transitive No.

CORRECTION Relations table - example 4. Cue: " aRb if $a = b = 1$ ". On $\mathbb{N}$ this relation is symmetric and not reflexive. The table again reverses those two entries. **Suggested revision:** Example 4 should read: symmetric Yes; reflexive No; transitive Yes.

CORRECTION Rational-number example. Cue: "On the set $Z \times N$ ". The intended domain is a Cartesian product. **Suggested revision:** Use $Z \times N$ (with the chosen convention $N = \{1, 2, \dots\}$, so denominators are non-zero).

CLARIFICATION Projection map. Cue: "The map $X \rightarrow X/R$ is called the projection map". The map is named but not defined. **Suggested revision:** Add $\pi: X \rightarrow X/R$, $\pi(x) = [x]$.

CLARIFICATION Distributive binary operations. Cue: " $A \cdot (B \sim C) = (A \cdot B) \sim (A \cdot C)$ ". This defines left distributivity only. In a non-commutative setting right distributivity is a separate condition. **Suggested revision:** Label the displayed law left distributivity and, if intended, add $(A \sim B) \cdot C = (A \cdot C) \sim (B \cdot C)$.

Page 9

CORRECTION Peano successor axioms. Cue: " S is injective (So we cannot loop back around...)". Injectivity alone does not prevent cycles and does not say that 1 is not a successor. **Suggested revision:** Add $1 \notin S(N)$ and avoid interpreting injectivity alone as the no-loop condition. State the induction axiom separately.

CORRECTION Convention for zero. Cue: "We define 0 ... If $c=0$ ". The section has defined N to start at 1, but the recursion and induction proof use 0 as though it were a natural-number base case. **Suggested revision:** Choose one convention consistently. With $N = \{1, 2, \dots\}$, prove associativity by induction from $c=1$; alternatively redefine N to include 0 and rewrite the successor axioms and earlier notation.

CORRECTION Associativity induction hypothesis. Cue: " $(a+b)+c=a+b+(b+c)$ ". The middle equality is false in general. **Suggested revision:** Delete the middle " $=a+b$ ", so the hypothesis is $(a+b)+c=a+(b+c)$. Also change $S(C)$ to $S(c)$ and cite the recursion equations at the steps where they are used.

CORRECTION Partial and total orders. Cue: "each pair satisfies exactly one ... [then] an ordering is Total if any two elements can be compared". The first condition is already trichotomy, hence totality; it is not satisfied by the subset partial order used as the example. **Suggested revision:** Define a partial order with $\leq$ as reflexive, antisymmetric and transitive. Add totality as: for all a, b , either $a \leq b$ or $b \leq a$. If using $<$, give the corresponding strict-order axioms.

CORRECTION Strong induction hypothesis. Cue: " $\{1, \dots, n\} \subseteq A$ implies $n+1 \in A$ ". A base condition $1 \in A$ is missing. Without it the empty set satisfies all displayed implications vacuously. **Suggested revision:** State: if $1 \in A$ and, for every $n \in N$, $\{1, \dots, n\} \subseteq A$ implies $n+1 \in A$, then $A = N$.

CLARIFICATION Proof that WPI implies SPI. Cue: "Apply WPI to the set of numbers n such that $\{1, \dots, n\} \subseteq A$ ". The idea is correct but the base and induction step should be shown, especially because the preceding SPI statement omitted its base case. **Suggested revision:** Let $B = \{n \in N : \{1, \dots, n\} \subseteq A\}$. Show $1 \in B$, then $n \in B$ implies $n+1 \in B$ by the SPI hypothesis; WPI gives $B = N$ and hence $A = N$.

Page 10

CORRECTION Well-ordering definition. Cue: "any subset has a least element". The empty subset has no least element. **Suggested revision:** Replace "any subset" with "every non-empty subset".

CORRECTION Relationship between WOP and SPI. Cue: "WOP does not imply SPI". For the usual natural numbers, WOP, weak induction and strong induction are equivalent. A least-counterexample proof derives SPI directly from WOP. **Suggested revision:** Replace this paragraph with: if A violates the conclusion of SPI, $N \setminus A$ has a least element m. The base case rules out $m=1$, and all smaller naturals lie in A, so the induction hypothesis forces $m \in A$, a contradiction. If a statement about more general successor structures is intended, announce that change of context explicitly.

CLARIFICATION Prime-factorization existence. Cue: "every number can be factored into primes". The statement should begin at 2, or adopt the convention that 1 is the empty product. **Suggested revision:** Use "every natural number $n \geq 2$ is a product of primes" and note that a composite least counterexample $n=ab$ has $1 < a, b < n$.

Page 11

CLARIFICATION Binomial-coefficient asymptotic. Cue: "When n is large, this is well approximated by $n^k/k!$ ". The approximation is valid as $n \rightarrow \infty$ with k fixed. It is not valid uniformly when k grows with n (for example $k=n$). **Suggested revision:** Add "for fixed k and $n \rightarrow \infty$ "; more precisely, $C(n,k) \sim n^k/k!$.

CLARIFICATION Inclusion-exclusion indexing. Cue: "the sums range over unordered tuples $(A,B,C,\dots)$ ". The indices A,B,C have not been defined as indices, and repeated indices must be excluded. **Suggested revision:** Write the standard formula using $1 \leq i_1 < \dots < i_r \leq n$, with sign $(-1)^{r+1}$, or sum over non-empty subsets $J \subseteq \{1, \dots, n\}$.

CLARIFICATION Indicator-function version. Cue: " $\sum i_{\{S_1 \cup \dots \cup S_n\}} = \dots$ ". A sum of indicators equals cardinality only after specifying the variable and a finite ambient set over which it is summed. **Suggested revision:** First state the pointwise identity for each x . Then, for finite sets, sum both sides over x in a stated finite ambient union.

Page 12

CORRECTION Composite numbers. Cue: "A composite number is a positive integer that is not prime". This makes 1 composite. **Suggested revision:** Use: "A composite number is an integer $n > 1$ that is not prime" (equivalently, $n=ab$ with $1 < a, b < n$).

CORRECTION Euclid's proof of infinitely many primes. Cue: "multiply them together and add 1 to get a new prime". The product plus one need not itself be prime; it has a prime divisor not in the original list. **Suggested revision:** Replace "get a new prime" with "obtain a number having a prime factor not in the list".

CLARIFICATION Division algorithm. Cue: "Let n and k be natural numbers". The divisor must be positive. The appeal to an integer and fractional part also imports a floor theorem rather than proving the result from the natural-number principles just developed. **Suggested revision:** State $k > 0$. For a first-principles proof, take the least q with $(q+1)k > n$ and put $r = n - qk$; then $0 \leq r < k$. Prove uniqueness by subtracting two representations.

CLARIFICATION Uniqueness of gcd. Cue: "the unique number dividing a and $b \dots$ ". Without positivity, both c and $-c$ have the divisibility property. **Suggested revision:** Define $\gcd(a,b)$ to be the unique positive common divisor divisible by every common divisor (with the usual separate convention for $\gcd(0,0)$, if needed).

Page 13

CORRECTION Linear-combination step in Bezout's proof. Cue: " $r=A-qh=A-(Ax+By)$ ". The second expression has lost the factor q . **Suggested revision:** Use $r=A-qh=A-q(Ax+By)=(1-qx)A+(-qy)B$.

CORRECTION Parity explanation in the Diophantine example. Cue: "*obviously no because the left hand side and the right hand side is odd*". The left-hand side $160x+72y$ is even (indeed divisible by 8), while 33 is odd. **Suggested revision:** Say: $\gcd(160,72)=8$ and 8 does not divide 33, so there is no integer solution.

Page 14

CLARIFICATION Irreducible versus prime in $\mathbb{Z}[i\sqrt{3}]$. Cue: "*2 and $1\pm i\sqrt{3}$ are 'prime' in this world*". They are irreducible, but not prime elements: 2 divides their product 4 without dividing either factor. That distinction is precisely what permits non-unique factorization. **Suggested revision:** Replace "prime" with "irreducible" and briefly note that irreducible need not imply prime in this ring.

CLARIFICATION Notation for the factors. Cue: " $2^{\{0\leq x\leq 3\}}3^{\{0\leq y\leq 7\}}11^{\{0\leq z\leq 1\}}$ ". Inequalities cannot serve as exponents. **Suggested revision:** Write $\{2^{x_1}3^{y_1}11^{z_1} : 0\leq x_1\leq 3, 0\leq y_1\leq 7, 0\leq z_1\leq 1\}$, with x,y,z integers.

CLARIFICATION Erdos counting bound. Cue: "*the k 'th prime must come before 4^k* ". The preceding contradiction establishes infinitude; the quantitative claim needs an explicit re-indexing. The underlying count gives $N\leq \sqrt{N}\cdot 2^{\pi(N)}$, hence $\pi(N)\geq (1/2)\log_2 N$. **Suggested revision:** Either omit the final sentence or derive the bound carefully from this inequality (and state the indexing/end-point convention).

Page 15

CORRECTION Definition of congruence. Cue: "*Alternatively, b is the remainder when a is divided by n* ". Congruent integers need not be the canonical remainder; for example $17\equiv 7 \pmod{10}$, but 7 is not the only integer congruent to 17. **Suggested revision:** Define $a\equiv b \pmod{n}$ by $n|(a-b)$. Only when $0\leq b<n$ may b be called the remainder of a upon division by n .

CORRECTION Addition and multiplication modulo n . Cue: " $a+b \bmod n=(a \bmod n)+(b \bmod n)$ ". As an equality of least non-negative remainders this is false without reducing the right-hand side again; $7+8$ gives $5 \bmod 10$, not 15. **Suggested revision:** Use $(a+b) \bmod n = ((a \bmod n)+(b \bmod n)) \bmod n$, and similarly for products; or state the cleaner congruence laws $a\equiv a', b\equiv b' \Rightarrow a+b\equiv a'+b'$ and $ab\equiv a'b'$.

Page 16

CLARIFICATION Number of solutions to $ax\equiv b$. Cue: "*the solution ... is exactly the solution to $(a/d)x\equiv b/d \pmod{n/d}$* ". When $d=\gcd(a,n)$ divides b , the reduced congruence has one class modulo n/d , corresponding to d classes modulo n . **Suggested revision:** State that $x\equiv x_0 \pmod{n/d}$; modulo n the solutions are $x_0, x_0+n/d, \dots, x_0+(d-1)n/d$. Thus $10x\equiv 12 \pmod{34}$ gives $x\equiv 8 \pmod{17}$, i.e. $x\equiv 8$ or $25 \pmod{34}$.

CLARIFICATION Chinese Remainder Theorem grammar. Cue: "*the simultaneous congruences ... has a unique solution*". The plural subject takes "have" and uniqueness is of a residue class modulo mn . **Suggested revision:** Use: "The simultaneous congruences have a unique solution modulo mn ."

Page 17

CLARIFICATION Multi-modulus CRT notation. Cue: " $x \equiv a_1 \pmod{m_1} \equiv a_2 \pmod{m_2} \equiv \dots$ ". The chain of equalities mixes different moduli and is not a single meaningful equality. **Suggested revision:** Display the system separately: $x \equiv a_i \pmod{m_i}$ for $i=1, \dots, k$.

CLARIFICATION Fermat proof. Cue: " $i-j < p$ ". The relevant statement is $|i-j| < p$, followed by $p \nmid (i-j)$ when $i \neq j$. **Suggested revision:** Use cancellation by the unit a , or write $0 < |i-j| < p$.

CORRECTION Totient of a product of primes. Cue: "If p and q are primes, $\phi(pq) = (p-1)(q-1)$ ". This is false when $p=q$. **Suggested revision:** Add that p and q are distinct primes. If $p=q$, $\phi(p^2) = p(p-1)$.

Page 18

CORRECTION Modulus in the converse proof. Cue: "Suppose there exists z such that $z^2 \equiv -1 \pmod{4}$ ". The assumed square root is modulo p , not modulo 4. **Suggested revision:** Replace " $\pmod{4}$ " with " $\pmod{p}$ ". Then, with $p=4k+3$, $z^{p-1} = z^{4k+2} = (z^2)^{2k+1} \equiv -1 \pmod{p}$, contradicting Fermat's little theorem.

Page 19

CLARIFICATION Carmichael-number wording. Cue: "always a false positive". A Carmichael number passes Fermat's test for every base coprime to it, not literally every integer base. **Suggested revision:** Add "for every base coprime to the number".

FACT CHECK Rank of the displayed generalized Fermat prime. Cue: "the 9358th largest known prime". This ranking is time-sensitive. PrimePages currently records the proven 551,525-digit prime as rank 10,147 and no longer in the Top 5,000 list. **Suggested revision:** Prefer a stable statement such as its digit count and discovery date, or date-stamp and update the rank. See the dated PrimePages record cited at the end of this review.

CLARIFICATION RSA security claim. Cue: "decrypting it is equally as hard as actually factoring n ". Factoring n is sufficient to recover $\phi(n)$ and the private exponent, but equivalence between breaking textbook RSA and factoring is not established in this general form. **Suggested revision:** Use: "No efficient general method is known for recovering the plaintext/private key from the public data at suitable sizes; the standard security assumption is closely related to the apparent difficulty of factoring n ."

CORRECTION Construction of $\mathbb{Q}$. Cue: " $\mathbb{Q}$ as the set of equivalence relations in $\mathbb{Z} \times \mathbb{N}$ ". The rationals are equivalence classes, not equivalence relations. **Suggested revision:** Replace "equivalence relations" with "equivalence classes".

Page 20

CLARIFICATION Rational square root argument. Cue: " $x = a^2/b^2 \dots$ so x must be square". The conclusion needs the fraction a/b in lowest terms, followed by comparison of prime exponents. **Suggested revision:** Add "write a/b in lowest terms" and use unique factorization to show $b=1$ and every prime exponent in x is even.

CORRECTION Ordered-field axioms. Cue: "transitive and antisymmetric (in the sense that $a < b$ and $b < a$ mean the same thing)". For a strict order, $a < b$ and $b < a$ are incompatible, not equivalent. The equivalent reversed notation is $a < b$ if and only if $b > a$. The field list also needs $0 \neq 1$. **Suggested revision:** State trichotomy and transitivity for $<$, which imply asymmetry; add $0 \neq 1$ to the field axioms.

CORRECTION Least-upper-bound property. Cue: "the least upper bound is in the set". A supremum need not belong to the set; the very next example gives $\sup(0,1)=1$ with $1 \notin (0,1)$. **Suggested revision:** Use: "Every non-empty $S \subseteq \mathbb{R}$ that is bounded above has a least upper bound $\sup S \in \mathbb{R}$." Then define it as an upper bound no larger than any other upper bound.

CLARIFICATION Construction versus characterization. Cue: "So here is how we construct the reals". The listed axioms characterize a complete ordered field up to isomorphism; they do not themselves construct one. **Suggested revision:** Change the lead to "We characterize $\mathbb{R}$ as a complete ordered field" or supply a genuine construction such as Dedekind cuts or Cauchy sequences.

Page 21

CORRECTION Peano axioms and integers. Cue: "the integers (as defined by the Peano axioms)". Peano axioms define the natural numbers, not the integers. **Suggested revision:** Use "the natural numbers defined by the Peano axioms, and the integers subsequently constructed from them".

CLARIFICATION Embedding $\mathbb{Q}$ in an ordered field. Cue: "everything is the successor of some element". That property describes the ambient field, not the embedded copy of $\mathbb{N}$, and it does not establish the claimed inclusions. **Suggested revision:** State that 1 generates a copy of $\mathbb{Z}$ under addition; $0 \neq 1$ and order imply characteristic 0; taking ratios embeds $\mathbb{Q}$ uniquely.

CLARIFICATION Infimum terminology and notation. Cue: "highest lower bound ... $\inf\{1/n\}=0$ ". The standard term is greatest lower bound, and $\inf$ should be applied to a set. **Suggested revision:** Write $\inf\{1/n : n \in \mathbb{N}\}=0$.

Page 22

CORRECTION Reciprocal Archimedean proof. Cue: " $(nt^{-1})/n < (nt^{-1})/y$ ". The variable y is undefined and the displayed comparison does not establish the claim. **Suggested revision:** After choosing $n > 1/t > 0$, take reciprocals to obtain $1/n < t$. If reciprocal order has not yet been proved, establish that positive reciprocation reverses inequalities.

CORRECTION Existence of $\sqrt{2}$. Cue: " $(c-t)^2 < c^2 - 4t + t^2 < c^2 - 3t$ ". Because $c < 2$, multiplying by $-2t$ reverses the relevant comparison: $-2ct > -4t$. Moreover, if the set A contains rational numbers only, $c+t$ need not lie in A . **Suggested revision:** Define $A = \{x \in \mathbb{R} : x \geq 0 \text{ and } x^2 < 2\}$. If $c^2 < 2$, choose small $t > 0$ with $(c+t)^2 < 2$. If $c^2 > 2$, choose small $t > 0$ with $(c-t)^2 > 2$; for instance bound $(c-t)^2 > c^2 - 4t$ and take $t < (c^2 - 2)/4$. Each case contradicts the defining property of c .

CLARIFICATION Rationality preservation sentence. Cue: "a number is rational iff adding it to or multiplying it by any non-zero rational gives a rational". The statement is true under a suitable quantifier reading but nearly tautological (take the rational 1), while the explanation proves only closure of $\mathbb{Q}$. **Suggested revision:** Either remove it or state the intended useful fact: rational $\pm$ rational and rational $\times$ rational are rational; rational plus irrational is irrational; non-zero rational times irrational is irrational.

Page 23

CORRECTION Quantifiers in the definition of convergence. Cue: "for any ϵ ... there is an N ... there exists a limit L ". The limit must be one fixed real number, chosen before ϵ ; n also needs a universal quantifier.

Suggested revision: Use: (a_n) converges to L if $L \in \mathbb{R}$ and $\forall \epsilon > 0 \exists N \in \mathbb{N} \forall n \geq N, |a_n - L| < \epsilon$. The sequence converges if such an L exists.

CORRECTION Divergence to positive infinity. Cue: "A sequence goes to infinity if $1/a_n \rightarrow 0$ ". This also holds for $a_n = -n$ and $a_n = (-1)^n n$, neither of which tends to $+\infty$. **Suggested revision:** Define $a_n \rightarrow +\infty$ by: for every $M \in \mathbb{R}$ there is N such that $n \geq N$ implies $a_n > M$.

CORRECTION Boundedness of a convergent sequence. Cue: "bounded by $\max(a_1, \dots, a_k)$ ". The definition uses $|a_n|$, so negative initial terms and the size of L must be included. **Suggested revision:** After taking $\epsilon = 1$, use $M = \max(|a_1|, \dots, |a_{N-1}|, |L| + 1)$.

Page 24

COPY EDIT Oscillating-sequence band. Cue: "a band of width less than -2 ". A width cannot be negative.

Suggested revision: Use "a band of total width less than 2" or simply take $\epsilon = 1/2$.

CORRECTION Symbolic negation of convergence. Cue: " $\exists \epsilon > 0 \forall N, L \dots k > N \not\Rightarrow \dots$ ". The limit variable, the term index and the quantifier order are all misplaced; implication is not negated by the symbol used here.

Suggested revision: Divergence is: $\forall L \in \mathbb{R} \exists \epsilon > 0 \forall N \in \mathbb{N} \exists k \geq N$ such that $|a_k - L| \geq \epsilon$.

CLARIFICATION Uniqueness of limits. Cue: "Let it be such that a and b are both limits". The proof should assume $a \neq b$ and choose one index large enough for both limit estimates. **Suggested revision:** Let $\epsilon = |a - b|/3$. Choose $N = \max(N_a, N_b)$; then $|a - b| \leq |a - a_n| + |a_n - b| < 2|a - b|/3$, a contradiction.

Page 25

CLARIFICATION Monotone convergence proof. Cue: "The least upper bound ... must be equal to its limit".

For a first-principles exposition this is the key step, not merely an assertion; decreasing sequences also require an infimum. **Suggested revision:** For an increasing bounded sequence set $L = \sup\{a_n\}$. Given $\epsilon > 0$, $L - \epsilon$ is not an upper bound, so some $a_N > L - \epsilon$; monotonicity gives $L - \epsilon < a_n \leq L$ for $n \geq N$. Treat decreasing sequences with the infimum.

CORRECTION Closed inequality under limits. Cue: "If $a_n \leq d$ for all d ". The quantified variable should be n .

Suggested revision: Use "If $a_n \leq d$ for all n and $a_n \rightarrow c$, then $c \leq d$." If $c > d$, take $\epsilon = (c - d)/2$; eventually $a_n > c - \epsilon > d$, a contradiction.

CLARIFICATION Epsilon bookkeeping. Cue: "we could have put ϵ ... and got 2ϵ ... that is ok". A proof for a requested tolerance ϵ should finish with a bound below that same ϵ . **Suggested revision:** Keep the $\epsilon/2$ argument, or say explicitly that because the tolerance is arbitrary one may rename 2δ as ϵ .

CORRECTION Geometric-series divergence. Cue: "blows up if $|r| \geq 1$ ". For $r = -1$ the partial sums oscillate and remain bounded; they do not blow up. **Suggested revision:** Say that for a non-zero first term the partial sums fail to converge when $|r| \geq 1$, with different behavior at $r = 1$, $r = -1$ and $|r| > 1$.

Page 26

CLARIFICATION Domain of the harmonic estimate. Cue: " $\ln(x) < \sum_{n=1}^x 1/n < 1 + \ln(x)$ ". The upper summation limit requires x to be an integer (or a floor). **Suggested revision:** State the estimate for integers $N \geq 1$: $\ln(N+1) \leq H_N \leq 1 + \ln N$, or retain the displayed strict version with its precise range.

CORRECTION Euler-Mascheroni expression. Cue: " $\sum_{n=1}^x 1/n - \ln(n)$ ". The free n in the logarithm should match the upper limit. **Suggested revision:** Use $H_N - \ln N$, which converges to the Euler-Mascheroni constant $\gamma \approx 0.57721$.

CORRECTION Grouped comparison for $\sum 1/n^2$. Cue: " $1 + 2/2^2 + 4/4^2 + 8/8^2 = \dots = 2$ ". The displayed finite expression equals 1.875, not 2; the equality to 2 needs all subsequent groups. **Suggested revision:** Insert $+\dots$ on the comparison side: $1 + 2/2^2 + 4/4^2 + 8/8^2 + \dots = 1 + 1/2 + 1/4 + 1/8 + \dots = 2$.

Page 27

CLARIFICATION Order of the exceptional decimal tails. Cue: "*the tails ... must be 00000... and 99999...*". Given $a_k < b_k$, the a -expansion must have the 9-tail and the b -expansion the 0-tail. **Suggested revision:** State the order explicitly and give the standard example $0.24999\dots = 0.25000\dots$.

CLARIFICATION Variables in the rational-periodic proof. Cue: " $x = a/b$... denominator $2^a 5^b q$ ". The letters a and b are reused both for the original fraction and for prime exponents. **Suggested revision:** Use $x = u/v$ in lowest terms and $v = 2^\alpha 5^\beta q$ with $\gcd(q, 10) = 1$.

Page 28

CORRECTION Lipschitz estimate. Cue: " $|p(x) - p(y)| \leq k(x - y)$ ". The right-hand side can be negative when $x < y$. **Suggested revision:** Use $|p(x) - p(y)| \leq K|x - y|$ for $x, y \in [0, 1]$.

CORRECTION Missing contradiction in the Liouville proof. Cue: "*non-zero integer multiples of $1/10^{-d} n!$ converging to 0 ... is impossible*". It is entirely possible: $10^{-d} n!$ itself is such a sequence and tends to zero. The proof needs incompatible lower and upper bounds. **Suggested revision:** For large n , $p(L_n) \neq 0$, so $|p(L_n)| \geq 10^{-d} n!$. But $p(L) = 0$ and the Lipschitz bound give $|p(L_n)| \leq K|L - L_n| \leq 2K \cdot 10^{-(n+1)!}$. Since $(n+1)! = (n+1)n!$ and $n+1 > d$, the upper bound is eventually smaller than $10^{-d} n!$, a contradiction.

CORRECTION General Liouville criterion. Cue: "*for all n there exists p/q within $1/q^n$ of x* ". As written every rational x satisfies the condition by taking $p/q = x$. **Suggested revision:** Require rational approximants $p/q \neq x$ with $q > 1$ (or infinitely many distinct approximants), and normally state the criterion for an irrational x .

Page 29

COPY EDIT Map in the subset lemma. Cue: "*the map $N \rightarrow s_n$* ". The codomain is a set, not a term depending on n . **Suggested revision:** Write $n \mapsto s_n$, a bijection $N \rightarrow S$ in the infinite case.

Page 30

CORRECTION Unique representation of rationals. Cue: "*lowest terms and $b \neq 0$* ". Both (a, b) and $(-a, -b)$ are then allowed, so the representation is not unique. **Suggested revision:** Require $b > 0$ as well as $\gcd(a, b) = 1$.

CLARIFICATION Countable union theorem. Cue: "For each A_i list its elements". Finite and empty A_i need handling, repeated elements in different A_i need a canonical representative, and simultaneous selection of enumerations invokes countable choice unless the listings are part of the data. **Suggested revision:** State the set-theoretic assumptions. With enumerations supplied, map x to the least encoded pair (i,j) for which $x=a_{ij}$; this gives an injection into $\mathbb{N} \times \mathbb{N}$.

CLARIFICATION Coefficient tuples. Cue: "degree k polynomials is exactly the same as $\mathbb{Z}^{k+1}$ ". The leading coefficient must be non-zero, so this is a subset rather than all of $\mathbb{Z}^{k+1}$. **Suggested revision:** Say it injects into $\mathbb{Z}^{k+1}$; that is enough for countability.

CLARIFICATION Decimal diagonal argument. Cue: "the number which differs ... is not in this list". Decimal non-uniqueness must be neutralized: a digitwise-different expansion can represent the same number through a $0.4999\dots = 0.5000\dots$ ambiguity. **Suggested revision:** Choose canonical expansions and construct the diagonal digit using only 1 and 2 (for example choose 1 unless the diagonal digit is 1, then choose 2). The constructed expansion cannot end in 9s and differs from row n in digit n .

Page 31

CORRECTION Bijection between $\mathbb{R}$ and $(0,1)$. Cue: " $\tan(x+\pi/2)$ ". On $x \in (0,1)$ this does not have range all of $\mathbb{R}$. **Suggested revision:** Use $x \mapsto \tan(\pi(x-1/2))$, which maps $(0,1)$ bijectively onto $\mathbb{R}$.

CORRECTION Decimal interleaving is not onto. Cue: "sending ... to $0.a_1b_1a_2b_2\dots$ is a bijection". Even after choosing canonical expansions, de-interleaving an arbitrary decimal can produce 0 or 1, or a component with an exceptional tail whose chosen representation no longer re-interleaves to the original. Surjectivity has not been proved. **Suggested revision:** Use Cantor-Schroeder-Bernstein: one injection is $x \mapsto (x, 1/2)$; for the reverse injection use a digit code with reserved blocks/separators and a fixed canonical expansion, or cite a proved standard encoding that handles exceptional tails.

Page 32

CORRECTION Binary expansion wording. Cue: "Consider the terminating binary expansion". Most numbers in $(0,1)$ do not have a terminating binary expansion. **Suggested revision:** Choose the canonical binary expansion that is not eventually all 1s (so dyadic rationals use their terminating 0-tail), then map its 1-positions to a subset of $\mathbb{N}$.

CORRECTION Conclusion that $\mathcal{P}(\mathbb{N})$ is uncountable. Cue: "if there was an injection from $\mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$ ". Such an injection certainly exists and gives no contradiction. **Suggested revision:** Once there is an injection $(0,1) \rightarrow \mathcal{P}(\mathbb{N})$, uncountability follows because if $\mathcal{P}(\mathbb{N})$ were countable it would inject into $\mathbb{N}$, and the composition would inject the uncountable interval into $\mathbb{N}$.

CORRECTION Disjoint open intervals. Cue: "We cannot add uncountably many numbers and get a finite number". The family is not assumed to lie in a bounded interval, so no finite total length has been established. The preceding rational-point argument is already sufficient. **Suggested revision:** Fix an enumeration of $\mathbb{Q}$ and assign to each interval the first rational it contains. Disjointness makes this assignment injective, so the family is countable. Remove the length argument.

CORRECTION Direction of the uncountability test. Cue: " X is uncountable if X has an injection to another uncountable set". $\mathbb{N}$ injects into $\mathbb{R}$, so this criterion would incorrectly make $\mathbb{N}$ uncountable. **Suggested revision:** Reverse it: if an uncountable set injects into X , then X is uncountable.

CLARIFICATION Injection-surjection equivalence. *Cue: "from a surjection ... pick something in the corresponding pre-image".* For arbitrary families of non-empty fibres this selection is an application of the axiom of choice. **Suggested revision:** State "assuming the axiom of choice" or restrict the lemma to a setting where a canonical choice is available. The injection $\Rightarrow$ surjection direction only needs A to be non-empty.

Page 33

CLARIFICATION Cantor-Schroeder-Bernstein proof. *Cue: "pick the element $g^{-1}(a)$... ancestor sequence".* The inverse symbols are partial, the parity classes are difficult to verify from their definitions, and the displayed h depends on claims that have not been cleanly established. **Suggested revision:** Use the standard construction. Given injections $f:A\rightarrow B$ and $g:B\rightarrow A$, set $A_0=A\setminus g(B)$, $A_{n+1}=g(f(A_n))$, and $A^*=\bigcup_{n\geq 0}A_n$. Define $h(a)=f(a)$ for $a\in A^*$ and $h(a)=g^{-1}(a)$ for $a\notin A^*$. Here g^{-1} is well-defined on $A\setminus A^*$ because those points lie in $g(B)$; a short check proves h bijective.

CLARIFICATION Final interval example. *Cue: "3x or something else".* This is a placeholder rather than a defined surjection. **Suggested revision:** Either rely only on the proved theorem after giving explicit injections both ways, or supply a complete piecewise map. For example, the identity injects $[0,1]$ into the union, while $f(x)=\sin^2(2\pi x)$ on $[0,1/2]$ and $f(x)=2+\sin^2(2\pi x)$ on $(1/2,1]$ is a surjection onto $[0,1]\cup[2,3]$.

Consolidated copy-edit sweep

These are mechanical edits that can be applied after the mathematical revisions. They are grouped to avoid repeating the same explanation in the page notes.

Page	Current form	Suggested form
1	ancient greeks / ancient greek times	ancient Greeks / ancient Greece
2	eachother; plain english; ie	each other; plain English; i.e.
3	, let's make a truth table	Let's make a truth table
4	venn; identites	Venn; identities
10	corrolary	corollary
11, 15, 16, 18, 26	Lets	Let's
13	conatins	contains
14	graetest; begininng; must me	greatest; beginning; must be
15, 26	eachother	each other
16	redidue; simaltaneous	residue; simultaneous
17	simaltaneous; Equivelently	simultaneous; Equivalently
19	equivelence	equivalence
22	highest lower bound	greatest lower bound

Page	Current form	Suggested form
23, 29	enumerated / enumerate	enumerated / enumerate
24	archimedian; simaltaneously; straight forward	Archimedean; simultaneously; straightforward
28	poylnomial	polynomial
29	inejction	injection
32	archimedian	Archimedean
Through out	ie / eg	i.e. / e.g. (or use 'that is' / 'for example')
Through out	hcf / Hcf / gcd and Lcm	Choose one capitalization and notation consistently

Items checked and retained

The following potentially suspicious points were checked and do not need a mathematical correction. They are recorded here to distinguish deliberate choices from oversights.

- The counterexample to ' $9 \mid n^2$ implies $9 \mid n$ ' is correct.
- The image and pre-image calculations for $x \mapsto x^2$ on the displayed open intervals are correct once pre-images are written as sets.
- The Euclidean-algorithm example $\gcd(420, 792) = 12$ is correct.
- The Bezout identity $1 = 3 \cdot 87 - 5 \cdot 52$ is correct, despite the crowded intermediate substitutions.
- The list 341, 561 and 645 is the standard list of base-2 Fermat pseudoprimes below 1000.
- The forward half of the criterion for -1 to be a quadratic residue mod p correctly uses Wilson's theorem when $p \equiv 1 \pmod{4}$.
- The construction $p + (q - p)/\sqrt{2}$ gives an irrational point strictly between rational $p < q$.
- The proof that e is irrational has the right factorial-tail structure; it mainly needs small notation and spacing edits.
- The map $\mathbb{R} \rightarrow \mathbb{R} \setminus \{0\}$ that shifts the non-negative integers and fixes every other real is a valid bijection.

External fact-check source

PrimePages entry for $260370566^{65536} + 1$ - current rank, digit count and verification status checked 30 July 2026: [PrimePages record](#)

Editorial note: no LaTeX command syntax, hyperlink-border rendering or editor-generated spacing has been treated as an authorial defect.